

Abstract :

The proposed automated saline level monitoring system leverages IoT and load cell technology to enhance patient safety and optimize resource utilization in healthcare settings. Traditional manual monitoring of saline levels is time-consuming, prone to errors, and inefficient. This system addresses these challenges by employing an HX711 load cell to measure the saline bottle's weight and transmitting the data to an IoT platform via MQTT for real-time monitoring. When the weight drops below a predefined threshold, the system automatically halts saline flow using a servo motor and triggers an audio alert via a buzzer. Additionally, users can control the servo motor asynchronously through MQTT commands. This solution ensures accurate and reliable saline level monitoring, minimizes the risks of saline shortages and reverse flows, and reduces manual intervention, thereby improving overall efficiency and patient care.

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Introduction:

The Saline Monitoring Project is an innovative IoT-based system designed to automate the monitoring and management of saline levels in healthcare environments. By leveraging components such as the ESP32 microcontroller, load cell, HX711 load cell amplifier, battery, buzzer, servo motor, and MQTT communication protocol, this system ensures real-time monitoring and efficient saline level management. MQTT (Message Queuing Telemetry Transport) facilitates reliable, lightweight communication between the system and the cloud for seamless data exchange.

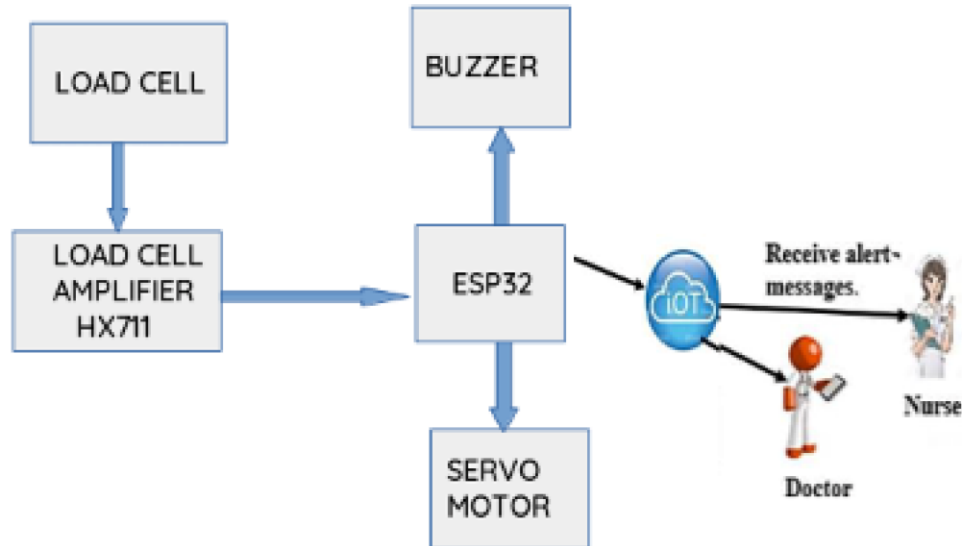
The project aims to eliminate the need for manual monitoring, reducing the risk of human error and enhancing operational efficiency. The ESP32 microcontroller acts as the central hub, collecting weight data from the load cell and HX711 sensor and wirelessly transmitting it to the MQTT server for real-time display and analysis.

The load cell, in combination with the HX711 amplifier, measures the weight of the saline bottle with high precision, providing accurate information about the saline level. This data is processed by the ESP32, enabling the system to trigger appropriate actions such as halting saline flow using a servo motor and alerting users with a buzzer when the saline level falls below a predefined threshold. Additionally, users can remotely control the servo motor via MQTT, ensuring adaptability and flexibility in system operation.

By automating saline level monitoring, this project significantly improves saline management, ensuring optimal usage and enhancing patient safety. It provides a reliable and efficient solution for healthcare facilities, reducing manual workload and ensuring timely responses to critical conditions.

System Design:

1. Proposed Model



2. Components

Hardware Components

- Load Cell (1kg): Measures the weight of the saline bottle.
- HX711: Amplifies and converts the load cell's analog signal to digital.
- ESP32: Facilitates communication with the MQTT server and controls peripherals.
- Servo Motor: Controls the flow valve based on saline level.
- Buzzer: Triggers an alert when the saline level is critically low.

Software Components

- Arduino IDE: For programming the ESP32.
- MQTT Broker (e.g., Mosquitto): Facilitates real-time data exchange between the system and the user.

3. Sensor Calibration

We calibrated the load cell and HX711 sensor to ensure precise and reliable measurement of saline levels. The load cell, which uses strain gauges to generate an electrical signal proportional to applied force, was connected to the HX711 module for signal amplification and Analog-to-Digital Conversion (ADC). During calibration, standard weights were used to map the raw output signal to accurate weight values, ensuring the system's measurements are meaningful and consistent for real-world application.

4. Cloud Data Transmission

Programmed the ESP32 to send weight data from the HX711 load cell amplifier to a cloud platform at regular intervals. The system ensures secure and reliable data transmission, enabling real-time storage and accessibility.

5. MQTT(Message Queuing Telemetry Transport) Broker Setup

Configured the laptop as a local MQTT broker using the open-source Mosquitto MQTT broker software. The broker manages the exchange of weight data and control commands between the ESP32 and connected devices within the same local network.

6. IP Address Sharing for Connectivity

Shared the laptop's IP address with other devices on the local network, allowing seamless access to MQTT topics. This enables multiple devices, such as smartphones, to monitor data and interact with the system remotely.

7. Real-Time Monitoring

Established a system where connected devices can access and visualize weight data in real time. This ensures timely updates, facilitating effective monitoring and prompt responses when saline levels drop below the threshold.

8. Two way Communication

Enabled two-way communication, allowing the ESP32 to publish weight data and subscribe to motor control commands. This supports remote operation of the servo motor for controlling the saline flow and updating system settings, ensuring adaptability and user control.

9. Threshold-Based Alerts

Predefined thresholds were set at 10 ml (warning) and 5 ml (critical) for saline levels. The ESP32 continuously monitors the data and triggers a buzzer alert when the weight falls below 10 ml, notifying users of the low saline level.

10. Automated Flow Control

If no action is taken after the buzzer alert, and the saline weight drops below 5 ml, the system automatically activates the servo motor to stop the saline flow, ensuring safe operation and preventing reverse flow.

11. User Control

The system allows the user to remotely control the servo motor via MQTT, providing asynchronous control to turn the motor on or off, deciding based on the published weight data, enabling manual intervention or adjustments as needed.

12. System Testing and Validation

Comprehensive testing was conducted to ensure the proper functionality and accuracy of the entire system. This included verifying the correct operation of all hardware components, including the load cell, HX711 amplifier, servo motor, and buzzer. The wireless communication and data acquisition processes were also tested to ensure reliable transmission of saline data. Additionally, the accuracy of the monitored saline levels was validated by comparing the system's readings against manual measurements to confirm the system's precision.

Working Principle:

1. The load cell measures the saline bottle's weight and sends the data to the HX711 module for amplification and analog-to-digital conversion.
2. The ESP32 reads the processed weight data from the HX711 and sends it to the MQTT cloud server for remote monitoring.
3. When the weight falls below a predefined threshold (e.g., 10% of the initial weight), the buzzer is activated to alert medical staff, and the servo motor halts the saline flow to prevent air embolism if no action is taken after buzzer alert.
4. Users can remotely control the servo motor via MQTT commands to override automatic actions, allowing manual intervention when necessary.

Results:

- Real-time weight data was successfully published to the MQTT server and displayed on a mobile application.
- The servo motor effectively stopped the saline flow at the threshold, and the buzzer provided timely alerts.
- Manual motor control through MQTT commands was responsive and reliable.

Conclusion:

In conclusion, the IoT-based saline monitoring system has successfully provided an efficient and reliable solution for real-time tracking of saline levels in healthcare settings. Using the ESP32 microcontroller and HX711 sensor, the system accurately measures and transmits data, while the automatic alert feature ensures timely intervention when saline levels drop below predefined thresholds. The system also offers remote control functionality, allowing healthcare professionals to manually intervene if needed.

Looking forward, future work will focus on designing a custom PCB to streamline the system's components and improve reliability. Additionally, upgrading to a high-torque servo motor will enhance control accuracy for better fluid regulation. The system could also be expanded to monitor various fluids beyond saline, broadening its applications in diverse medical settings and other industries requiring precise fluid management.

References :

- [1] "IoT-Based Saline Level Monitoring and Control System for Healthcare". Authors: Gupta, S., & Chaurasia, V. Published: 2020.
- [2]. "Smart Saline Monitoring System Using IoT and Cloud Computing". Authors: Patel, K., & Singh, R. Published: 2019.
- [3]. "Design and Implementation of an IoT-Based Saline Infusion Monitoring System". Authors: Arora, M., et al. Published: 2018.